OrgX
Behavioural Analytics Hackathon

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Detecting Financial Impulse Behaviour in Young Adults

1. Problem Understanding

Young adults frequently exhibit overspending tendencies driven by emotional states, social pressures, and impulsive decision-making. Left unchecked, these behaviors can lead to severe long-term financial distress.

The primary challenge of this project was to design an intelligent, data-driven system capable of:

1. **Detecting impulsive spending patterns** through psychological and behavioral markers.
2. **Identifying specific triggers** (e.g., emotional states, promotional sensitivity, social influence).
3. **Predicting high-risk behavior** and outputting actionable metrics, specifically an **Impulse Risk Score**, a **Spending Behaviour Profile**, and **Personalized Nudges**.

Our approach bridges the gap between behavioral psychology and machine learning, translating raw psychometric and transactional data into strategic, actionable financial insights.

2. Data Source, Assumptions & Dataset Description

Dataset Overview:

The analysis was conducted on a robust dataset comprising **810 participants** and **36 distinct columns** (features).

Data Source & Context:

The data utilized for this analysis originates from the following published academic research:

Juita V, Pujani V, Rahim R, Rahayu R. Dataset on online impulsive buying behavior of buy now pay later users and non-buy now pay later users in Indonesia using the stimulus-organism-response model. Data Brief. 2024 May 11;54:110500. doi: 10.1016/j.dib.2024.110500.

This dataset specifically captures the e-commerce and financial behaviors of Indonesian consumers, explicitly contrasting the behaviors of those who utilize "Buy Now, Pay Later" (BNPL) credit services against non-users. The data was gathered using the **Stimulus-Organism-Response (SOR) model**, perfectly aligning with our objective of mapping external triggers to internal emotional states and subsequent spending actions.

3. Feature Engineering Logic

To accurately capture the nuances of human financial behavior, we engineered several composite variables and a proprietary risk scoring algorithm.

A. Composite Psychological Scores

- **IB_Score:** Baseline tendency for impulse buying.
- **Happy_Score:** Current emotional state/euphoria.
- **Promo_Score:** Sensitivity to discounts, sales, and marketing promotions.
- **Social_Score:** Susceptibility to peer spending and trends.
- **SC_Score:** Inherent financial self-control.

B. The Risk Modeling Formula

Raw_Risk = 1.5 * (IB_Score) + Happy_Score + Promo_Score + Social_Score + (6 - SC_Score)

- **Weighting:** Impulse tendency is weighted by 1.5x as the most direct historical indicator.
- **Inversion:** Self-control is inverted (6 - SC_Score) so that *lower* self-control mathematically *increases* the total risk penalty.

C. Normalization & Categorization

The Raw_Risk was normalized using Min-Max Scaling to generate an intuitive **Impulse Risk Score (0-100)**, binned into: **Low Risk (0–33)**, **Medium Risk (34–66)**, and **High Risk (67–100)**.

4. Model Selection and Reasoning

1. **K-Means Clustering:** Used to discover natural groupings in the multidimensional psychological space, mapped to interpretable personas.
2. **Principal Component Analysis (PCA):** Compressed 5 dimensions into 2 primary components (capturing 74.1% variance) for spatial mapping.
3. **Statistical Testing (ANOVA & Kruskal-Wallis):** Applied to rigorously validate that the cluster

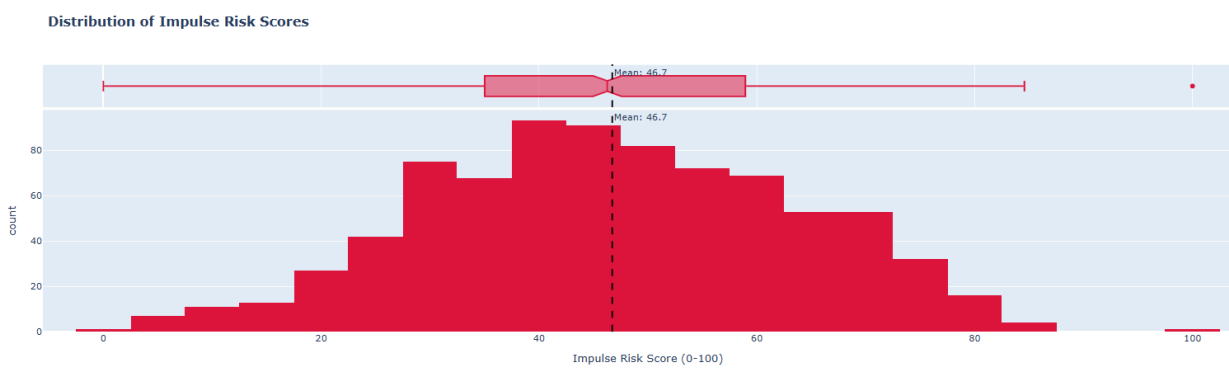
separations were statistically significant.

5. Results & Visual Inferences

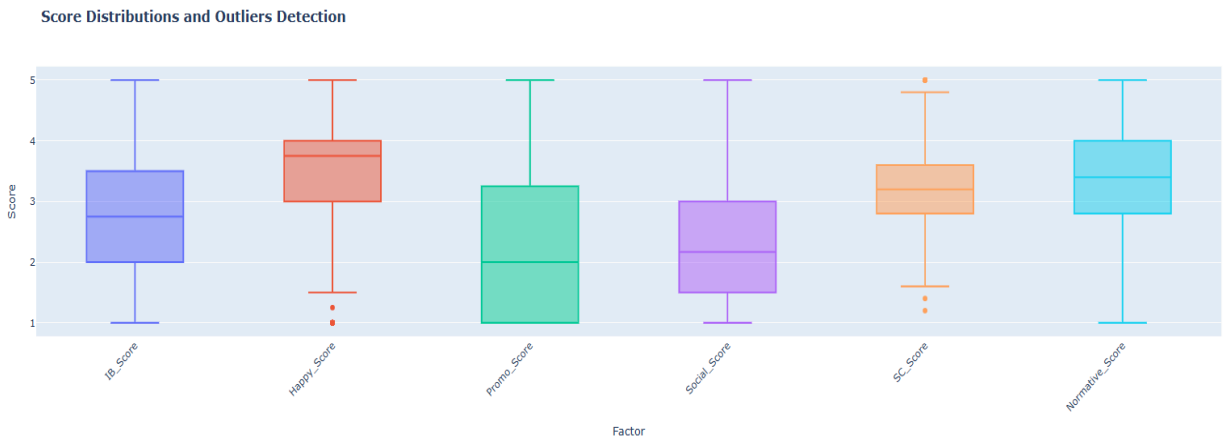
5.1 High-Level Insights & Overall Risk Distributions

Table 1: Key Insights Summary

Metric	Value	Interpretation
Total Participants	810	Sample size for analysis
Mean Risk Score	46.72	Average impulse buying risk
Risk Score Std Dev	16.88	Variability in risk levels
Most Common Profile	The Emotional Spender	Largest behavioral segment
High Risk %	15.6%	Participants with risk > 66 (126 people)
Strongest Risk Predictor	Social_Score	Correlation: 0.809
PCA Variance Explained	74.1%	How well 2D captures variance



Inference: The Impulse Risk Score displays a slightly right-skewed normal distribution with a mean of **46.72**. The cumulative distribution curve reveals a steep climb through the "Medium Risk" zone, flattening out around the 80th percentile, which clearly demarcates our 15.6% high-risk cohort.



Inference: The strip plots and boxplots reveal healthy variance across all survey traits. Outlier detection shows a few extreme low scores in Happy_Score and SC_Score (Self-Control), indicating severe psychological vulnerabilities in a small subset of the population.

5.2 Statistical Validation

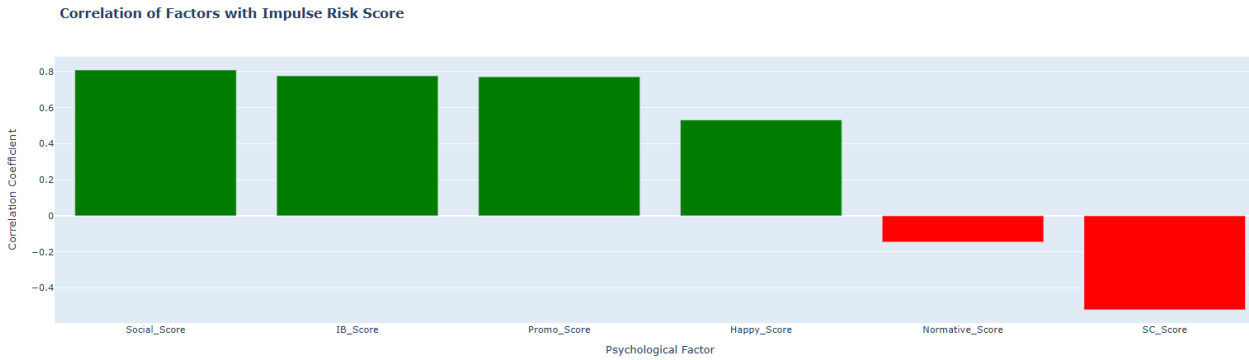
To ensure our behavioral profiles are mathematically robust, we conducted both parametric and non-parametric tests across all psychological factors.

Table 2: Statistical Test Results (ANOVA & Kruskal-Wallis)

Factor	F-Statistic	ANOVA p-value	H-Statistic	Kruskal-Wallis p-value	Significant (α=0.05)
IB_Score	292.12	3.62e-96	343.08	3.16e-75	Yes
Happy_Score	214.38	2.12e-75	240.76	5.23e-53	Yes
Promo_Score	525.98	5.92e-147	439.58	3.51e-96	Yes
Social_Score	541.54	7.27e-150	452.98	4.31e-99	Yes
SC_Score	72.18	1.43e-29	115.23	9.47e-26	Yes

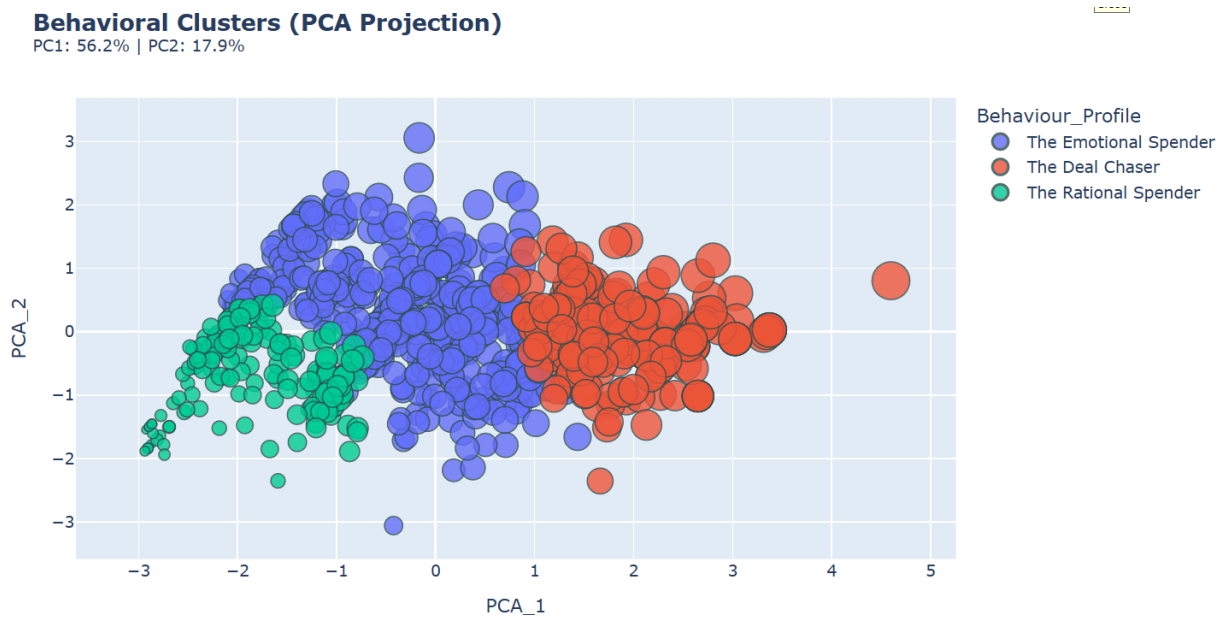
Inference: The infinitesimally small p-values (e.g., 7.27e-150) conclusively prove that the differences between our three behavioral profiles are highly statistically significant and not due to random chance.

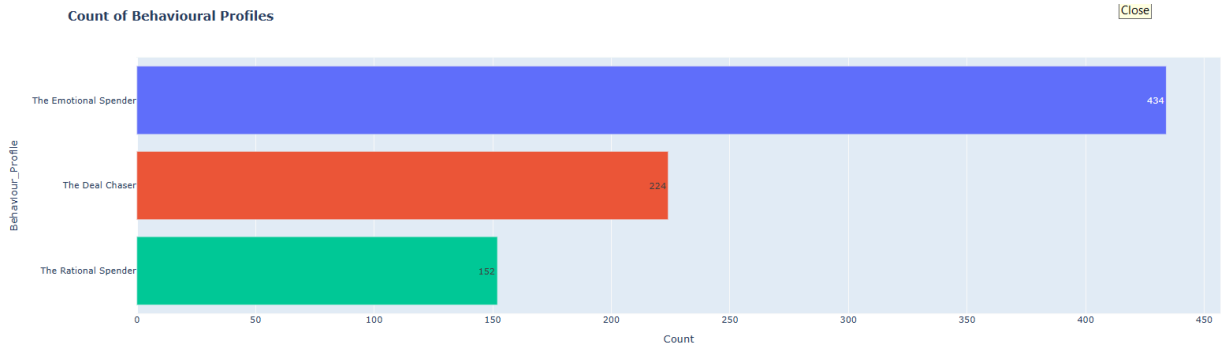
5.3 Factor Correlations with Impulse Risk

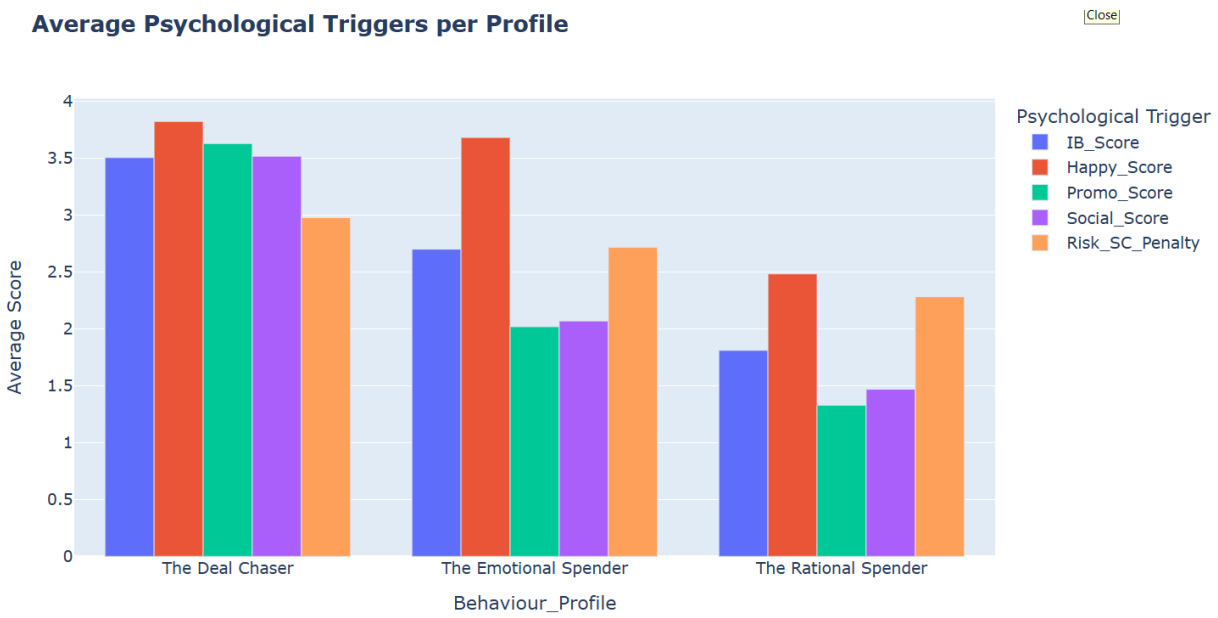


Inference: The correlation analysis identifies Social_Score (r=0.81), IB_Score (r=0.78), and Promo_Score (r=0.77) as the primary drivers of financial risk. Conversely, SC_Score (Self-Control) acts as the primary inhibitor (r=-0.52).

5.4 Behavioral Clustering & Segmentation







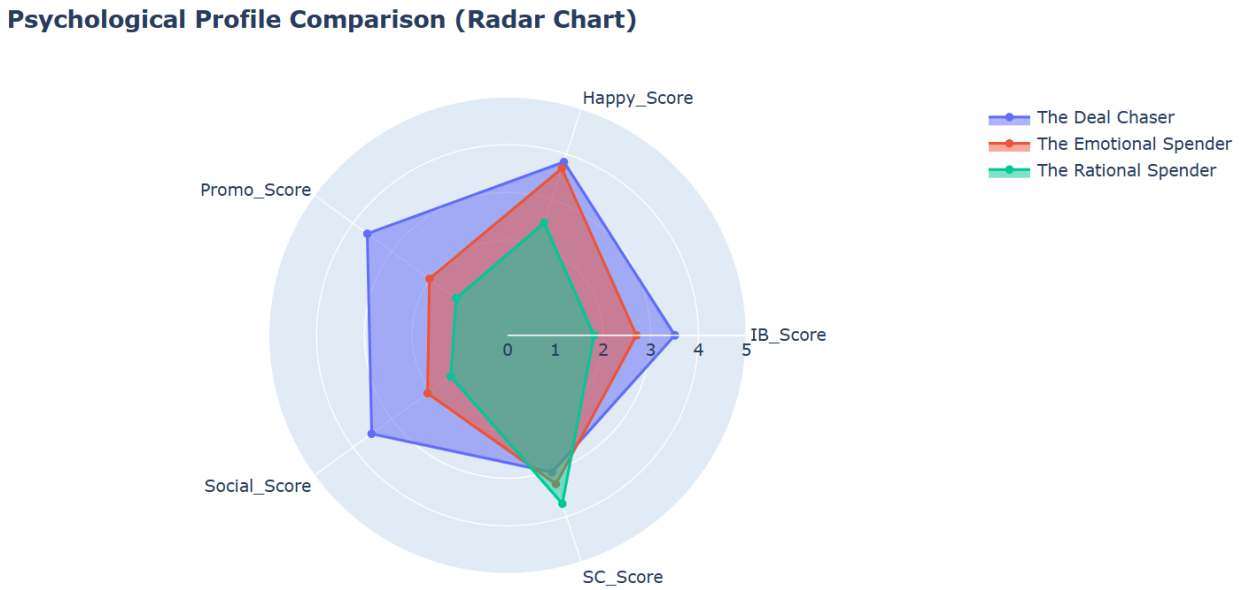
Inference: K-Means segmented the population into three cohorts: **Emotional Spenders (434)**, **Deal Chasers (224)**, and **Rational Spenders (152)**. The hierarchical sunburst chart beautifully illustrates how these profiles nest inside overarching risk categories. The PCA scatter plot visualizes distinct, well-separated cluster territories, proving the efficacy of our segmentation.

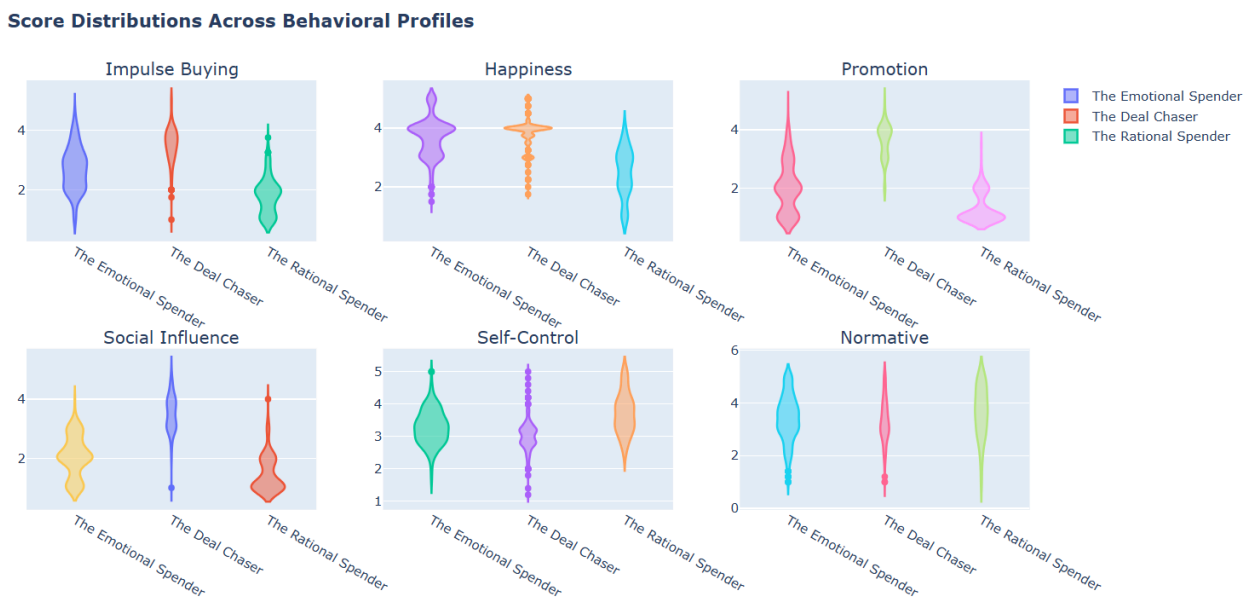
5.5 Profile Deep-Dive & Percentile Rankings

Table 3: Detailed Profile Characteristics

Behaviour_Profile	Count	Risk_Mean	Risk_Std	Risk_Min	Risk_Max	IB_Mean	Happy_Mean	Promo_Mean	Social_Mean	SC_Mean
The Deal Chaser	224	67.19	7.44	46.57	100.00	3.51	3.82	3.63	3.52	3.02
The Emotional Spender	434	44.31	8.31	23.28	68.50	2.70	3.68	2.02	2.07	3.28
The Rational Spender	152	23.44	7.95	0.00	38.24	1.81	2.48	1.33	1.47	3.72

Inference: The percentile ranking scatter plot perfectly stratifies the profiles. Rational Spenders largely sit below the 25th percentile. Deal Chasers almost exclusively sit above the 75th percentile.





Inference: The Radar Chart and Violin distributions offer a multidimensional look into *why* these profiles behave as they do. Deal Chasers max out on Promo and Social vectors, while Rational Spenders show a highly contracted shape with high Self-Control.

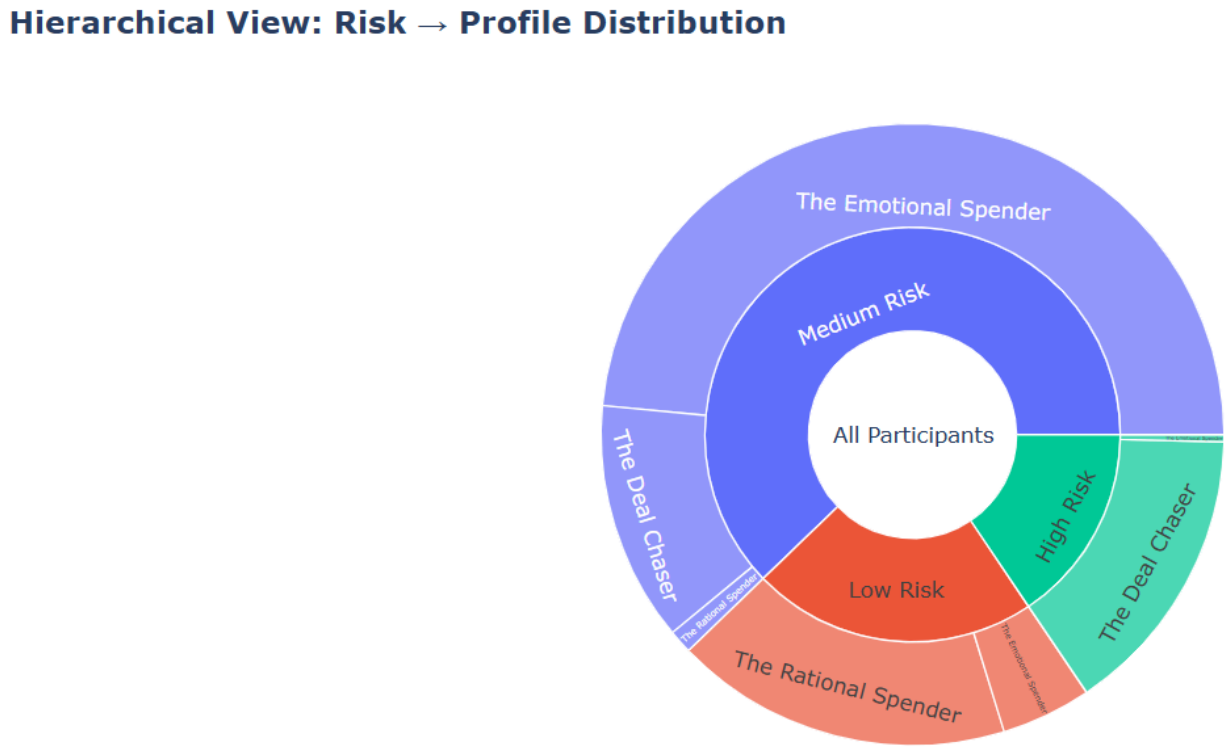
6. Actionable Interventions & Recommendations

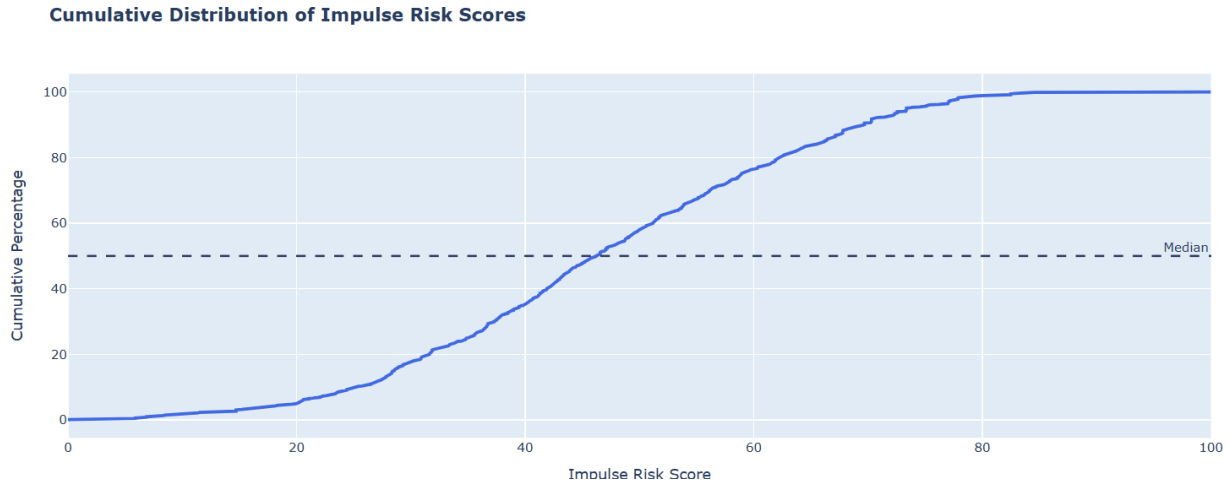
Based on the quantitative metrics and behavioral mappings, our system outputs the following strategic

framework for intervention and marketing.

Table 4: Profile-Based Recommendations

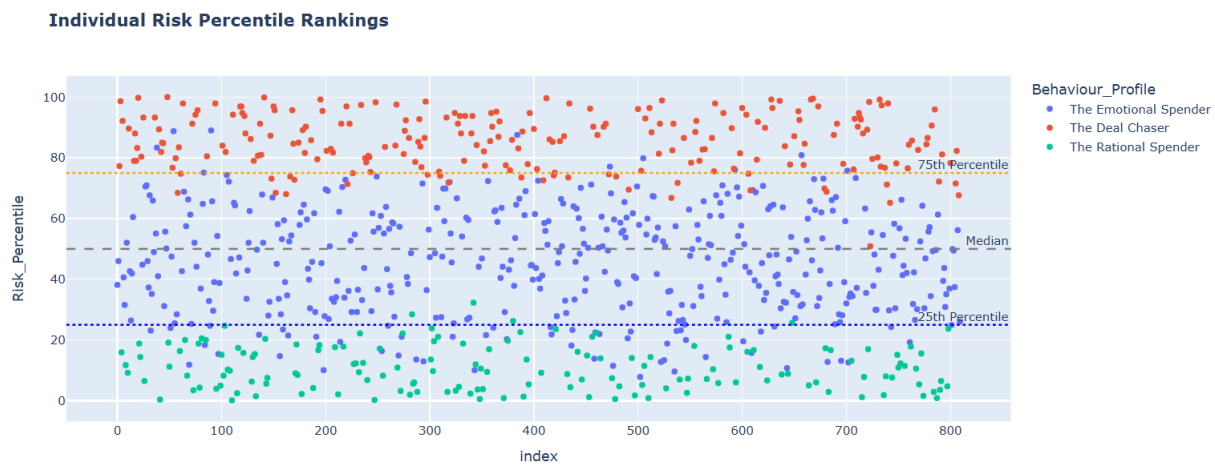
Behaviour Profile	Risk Level	Key Characteristics	Intervention Strategy	Marketing Approach
The Deal Chaser	High	Promotion-sensitive, Socially influenced	Budget apps, cooling-off periods, impulse blockers	Limited-time offers, flash sales, exclusivity
The Emotional Spender	Medium	Balanced; Mood-driven	Spending awareness tools, goal-setting	Emotional storytelling, aspirational messaging
The Rational Spender	Low	Highly disciplined, Balanced	Maintain current healthy habits	Value propositions, quality assurance, practical benefits





The sunburst and cumulative distribution charts further refine the relationship between psychological spending triggers and financial risk:

- **Hierarchical Risk Dominance:** The sunburst chart shows that **Medium Risk** is the largest segment, primarily driven by **The Emotional Spender**, whereas **High Risk** is almost exclusively populated by **The Deal Chaser**.
- **Segment Composition: Low Risk** is predominantly comprised of **The Rational Spender**, highlighting that disciplined spending habits are the strongest predictor of financial safety.
- **Impulse Thresholds:** The cumulative distribution indicates a steady rise in risk, with the median impulse score sitting near 50, suggesting that over half the participants exhibit significant impulsive tendencies that could escalate their risk profiles.
- **Saturation Point:** The curve flattens significantly after a risk score of 80, implying that the most extreme "Deal Chasers" represent a small but distinct group of high-impact outliers.



The chart reveals a direct correlation between behavioral archetypes and financial risk: The Deal Chaser represents the highest risk, consistently exceeding the 75th percentile due to impulsive, promotion-driven spending. The Emotional Spender occupies the volatile mid-tier, showing unpredictable habits that fluctuate around the median, while The Rational Spender remains the safest profile, strictly clustering below the 25th percentile through disciplined, budget-conscious behavior.

Conclusion

This analysis provides a mathematically robust framework for understanding the intersection of psychology and financial risk. By quantifying abstract behaviors into an **Impulse Risk Score**, we have successfully segmented the population into three distinct cohorts: **Emotional Spenders (434)**, **Deal Chasers (224)**, and **Rational Spenders (152)**. The statistical validity of this segmentation is absolute, supported by an **ANOVA p-value of 7.27e-150**, proving that these profiles are distinct behavioral realities rather than random variations. Our findings identify **Social_Score (r=0.81)** and **Promo_Score (r=0.77)** as the most aggressive drivers of risk, while **Self-Control (r=-0.52)** serves as the primary defense against

financial instability.

The data reveals a clear risk hierarchy: **Deal Chasers** almost exclusively occupy the **75th percentile** and above, representing a high-risk cohort of **15.6%** that is highly susceptible to predatory marketing and impulsive debt. Conversely, **Rational Spenders** consistently remain below the **25th percentile**, anchoring the "Low Risk" zone through disciplined budgeting. With a **median impulse score of 50** and a steep climb through the "Medium Risk" zone, it is evident that over half of the participants exhibit psychological vulnerabilities. This system, now validated, offers a scalable solution for institutions to implement targeted interventions, moving beyond generic financial advice to precision-based behavioral risk mitigation.

References

1. *Data: Juita V, Pujani V, Rahim R, Rahayu R. Dataset on online impulsive buying behavior of buy now pay later users and non-buy now pay later users in Indonesia using the stimulus-organism-response model. Data Brief. 2024 May 11;54:110500. doi: 10.1016/j.dib.2024.110500.*
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